Complete proof from last time

(vii) $\Rightarrow$ (i) (i.e. $\|Ux\|_2 = \|x\|_2$, $\forall x \Rightarrow U$ unitary)

<u>Lemma</u> If $A \in M_n$ and $x^* A x = 0$, $\forall x \in \mathbb{C}^n$, then $A = 0$ (zero matrix)

<u>Proof of lemma</u> By clever choice of x:

choose $x = e_p + i e_q$

where e_p, e_q are the p-th & q-th columns of $I \in M_n$, $i = \sqrt{-1}$

Then

$$0 = x^* A x = (e_p^T - i e_q^T) A (e_p + i e_q)$$

$$= \cancel{e_p^T A e_p}^{\,0} + e_p^T A (i e_q) - i e_q^T A e_p + \cancel{e_q^T A e_q}^{\,0}$$

$$\Rightarrow \boxed{e_p^T A e_q = e_q^T A e_p} \quad (1)$$

$\Rightarrow$ $(p,q)^{th}$ entry of A = $(q,p)^{th}$ entry of A

(2)

Now choose $x = e_p + e_q$

$$0 = x^* A x = e_p^T A e_q + e_q^T A e_p \Rightarrow \boxed{e_p^T A e_q = -e_q^T A e_p}$$

By 1 & 2 $\begin{cases} p \neq q \text{ (off-diagonal entries)} = 0 \\ p = q \text{ of course} = 0 \end{cases}$

Thus $A = 0$. $\square$

Back to showing (vii) $\Rightarrow$ (i).

Assume $\|x\|_2 = \|Ux\|_2 \quad \forall x \in \mathbb{C}^n$.

$$\Rightarrow \quad x^* x = (Ux)^* (Ux)$$

$$\Rightarrow \quad x^* x = x^* U^* U x, \quad \forall x \in \mathbb{C}^n$$

$$\Rightarrow \quad x^* (I - U^* U) x = 0, \quad \forall x \in \mathbb{C}^n$$

$$\xRightarrow[A = I - U^* U]{\text{Lemma}} \quad I - U^* U = 0 \Rightarrow U^* U = I$$

$\Rightarrow$ U is unitary $\blacksquare$

Fun questions

(1) What can you say about the eigenvalues of a unitary matrix U?

$$Ux = \lambda x \ , \ x \neq 0 \ , \ \lambda \in \sigma(U)$$

Then $\underbrace{\|Ux\|_2 = \|x\|_2}_{(vii)}$, $\|Ux\|_2 = \|\lambda x\|_2$

So $\|x\|_2 = \|\lambda x\|_2 = |\lambda| \, \|x\|_2 \ , \ x \neq 0$

$$\overset{\|x\|_2 \neq 0}{\Longrightarrow} \quad |\lambda| = 1$$

i.e. eigenvalues of U live on the unit circle in $\mathbb{C}$.

<u>Comment</u> Unfortunately this does not make find $\sigma(U)$ any easier.
(deep result by Watkins)

(2) What about the determinant of unitary U?

$$|\det U| = \left| \prod_{\lambda \in \sigma(A)} \lambda \right| = \prod_{\lambda \in \sigma(A)} |\lambda| = 1$$

<u>CHALLENGE</u> Can you discover (and then prove) other properties of unitary matrices?

Experiment → create lots of unitary matrices
↓ & compute/check
matlab

- eigenvectors
- diagonalizability
- characteristic polynomials (elem. symm. functions of the eigenvalues)

<u>How do you create examples</u>:

a) - diag. matrices $U = \mathrm{diag}(u_1, u_2, \ldots, u_n)$ with $|u_j| = 1$ $j = 1, 2, \ldots, n$

- We know $U = [u_1 | u_2 | \ldots | u_n]$ is unitary iff $\begin{cases} u_i^* u_j = 0, & i \neq j \\ u_i^* u_i = 1 & i = 1, 2, \ldots, n \end{cases}$

> So start with a linearly independent set
$$\{x_1, x_2, \ldots, x_n\} \subseteq \mathbb{C}^n$$
(eg columns of any invertible matrix)

> Apply Gram-Schmidt orthogonalization process
$$\tilde{u}_1 = x_1$$
$$\tilde{u}_2 = x_2 - \frac{x_2^* x}{x_1^* x_1} x_1$$
$$\vdots$$

Creates:
$$\{\tilde{u}_1, \tilde{u}_2, \ldots, \tilde{u}_n\}$$
an orthogonal set (still a basis for $\mathbb{C}^n$)

Normalize
$$u_j = \frac{\tilde{u}_j}{\|u_j\|_2}$$

Then $\{u_1, u_2, \ldots, u_n\}$ is orthonormal set

The $U = [u_1 | u_2 | \ldots | u_n]$ is unitary

(b) Quick alternative to find unitary matrices.

<u>Famous Theorem</u> (later)

Let $A \in M_n$. Then there exist unitary matrix Q and upper triangular matrix R such that $A = QR$

("QR-factorization" theorem)

Matlab command $[Q, R] = qr(A)$

$A = QR$ as above

keep $Q \rightarrow$ it is unitary!

Special types of unitary matrices

- 2×2 real unitary matrices ("real orthogonal")

$$U \in M_2(\mathbb{R}), \quad U^T U = I$$

$$U = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad U^T U = \begin{bmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Enforce:

$$\begin{cases} a^2+b^2 = 1 \\ ac+bd = 0 \\ -1 \le a,b,c,d \le 1 \end{cases}$$

(lengths = 1 of columns)

Trigonometry!

$\exists \theta$ such that $a = \cos\theta \Rightarrow b^2 = \sin^2\theta = 1-\cos^2\theta$

$\Rightarrow b = \begin{cases} \sin\theta \\ -\sin\theta \end{cases}$. The other constraints are also satisfied (check) So

$$U = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

for some $\theta \in \mathbb{R}$.

Rotation (counterclockwise) by θ

2/1/23

(We'll come back to special types of unitary matrices)

Observations

(1) Let $U, V \in M_n$ be unitary. Then

$$(UV)^*(UV) = V^* U^* U V$$
$$= V^* I V = V^* V = I$$

i.e. UV is unitary!

This means that the set of unitary matrices in M_n form a multiplicative group (subgroup of the invertible matrices in M_n)

- closed under multiplication
- every element is invertible
- contains identity
- associative group

$$(UVW) = (UV)W$$
$$= U(VW)$$

(2) The set of unitary matrices in M_n is also topologically well-behaved.

Let $U \in M_n$ unitary, $U = [u_{ij}]$

$$U = [u_1 | u_2 | \cdots | u_n]$$

- Since every column has length 1, it follows that

$$|u_{ij}| \leq 1$$

$$\left(1 = \|u_j\|_2^2 = u_j^* u_j = \sum_{i=1}^{n} \overline{u}_{ij} u_{ji} = \sum_{i=1}^{n} |u_{ij}|^2 \right)$$

This results to U being a $\boxed{\text{bounded}}$ operator (i.e. the collection of all unitary matrices is a $\boxed{\text{bounded}}$ set.)

Also, the limit of a convergent sequence of unitary matrices must be also be unitary $\lim_{k\to\infty} U^{(k)} = U$, $U^* U = \lim_{k\to\infty} U^{(k)*} U^{(k)}$

$$= \lim_{k\to\infty} I = I$$

So collection of unitary matrices in M_n is $\boxed{\text{closed}}$

So collection of unitary matrices in M_n (which is isomorphic to $\mathbb{C}^{n^2}$) is closed & bounded, i.e. <u>compact</u>.

This has a practical implication:

<u>Lemma</u> (Selection Principle)

Let $U_1, U_2, \ldots$ be an infinite sequence of unitary matrices. Then there exist a subsequence

$$U_{k_1}, U_{k_2}, \ldots$$

such

$$\lim_{j \to \infty} U_{k_j} = U_0$$

U_0 is unitary.

<u>Proof</u> Just a property of compact sets.

<u>Example</u> $U_k = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^k$

$U_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $U_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $U_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \ldots$

All unitary.

Contains $U_1, U_3, U_5, \ldots \to \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

§ 2.2 Unitary equivalence

Definition $B \in M_n$ is unitarily equivalent to $A \in M_n$ if there exists unitary $U \in M_n$ such

$$B = U^* A U$$

(clearly an equivalence relation)

(clearly it is a similarity) $U^* = U^{-1}$

What else?

Theorem If $A = [a_{ij}]$, $B = [b_{ij}] \in M_n$ are unitarily equivalent, then

$$\sum_{i,j=1}^{n} |a_{ij}|^2 = \sum_{i,j=1}^{n} |b_{ij}|^2$$

(later: $\sum_{i,j=1}^{n} |a_{ij}|^2$ = matrix norm $= \|A\|_F$ = Frobenius norm of A)

Proof

We'll use the following fact:

$$A \in M_n, \quad A = [a_{ij}]$$

$$\operatorname{trace}(A^* A) = \sum_{i,j=1}^{n} |a_{ij}|^2$$

e.g.

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^* \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$= \begin{bmatrix} \bar{a}_{11} & \bar{a}_{21} \\ \bar{a}_{12} & \bar{a}_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$= \begin{bmatrix} \bar{a}_{11} a_{11} + \bar{a}_{21} a_{21} & * \\ * & \bar{a}_{12} a_{12} + \bar{a}_{22} a_{22} \end{bmatrix}$$

$$\Rightarrow \operatorname{trace}(A^* A) = |a_{11}|^2 + |a_{21}|^2 + |a_{12}|^2 + |a_{22}|^2$$

$$(z\bar{z} = |z|^2)$$

So, if $B = U^* A U$, U = unitary, then

$$\sum_{i,j} |b_{ij}|^2 = \operatorname{trace}(B^* B) = \operatorname{trace}(U^* A^* U U^* A U)$$

$$= \operatorname{trace}(U^* A^* A U)$$

$$= \sum_{\lambda \in \sigma(U^* A^* A U)} \lambda$$

But

$$\sigma(U^* A^* A U) = \sigma(A^* A)$$

(similar since $U^* = U^{-1}$)

So

$$\sum_{i,j} |b_{ij}|^2 = \sum_{\lambda \in \sigma(A^*A)} \lambda = \operatorname{trace}(A^* A)$$

$$= \sum_{i,j} |a_{ij}|^2 \quad \blacksquare$$

<u>Consequence</u> Unitary equivalences are preferable similarities, when possible. because the magnitudes of the entries are under control.

<u>Example</u> Let $A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$

Show A, B are similar, but not unitarily similar.

$\sigma(B) = \{1, 2\}$

$\lambda, \mu \in \sigma(A)$: $\left.\begin{array}{l} \lambda + \mu = 3 + 0 = 3 \\ \lambda \cdot \mu = \det A = 2 \end{array}\right\} \Rightarrow \begin{array}{l} \lambda = 1 \\ \mu = 2 \end{array}$

$\sigma(A) = \{1, 2\}$

Distinct so both A, B are diag/ble

$A \sim \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

$B \sim \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

So $A \sim B$

Not unitarily similar:

$\sum_{i,j} |a_{ij}|^2 = 3^2 + 1^2 + (-2)^2 + 0^2 = 14$

$\sum_{i,j} |b_{ij}|^2 = 1^2 + 1^2 + 0^2 + 2^2 = 6$

$6 \neq 14$ Apply last theorem.

So keep in mind:

Unitary similarity (equivalence) $\Longrightarrow$ similarity

$\not\Longleftarrow$

(last example)

2/3/23

Finish § 2.2 :

Two concrete constructions of unitary matrices of specific utility (esp. in numerical linear algebra).

(1) Plane rotation / aka Givens rotations

$$U(\theta; i,j) = \begin{array}{c} \\ \\ \\ \\ \\ \\ \\ \end{array}\begin{array}{c} \begin{bmatrix} 1 & & & & & & & & & \\ & \ddots & & & & & & & & \\ & & 1 & & & & & & & \\ & & & \cos\theta & \cdots & & -\sin\theta & & & \\ & & & \vdots & \ddots & & \vdots & & & \\ & & & & & 1 & & & & \\ & & & \sin\theta & \cdots & & \cos\theta & & & \\ & & & & & & & \ddots & & \\ & & & & & & & & 1 & \end{bmatrix} \\ \begin{array}{ccccccccc} & & & \uparrow & & & \uparrow & & \end{array} \\ \text{i-th column} \qquad \text{j-th column} \end{array} \begin{array}{l} \\ \\ \\ \leftarrow \text{i-th row} \\ \\ \\ \leftarrow \text{j-th row} \\ \\ \\ \end{array}$$

$\theta \in [0, 2\pi]$

e.g.

$U = U(\theta; 1,3) \in M_4$

$$U = \begin{bmatrix} \cos\theta & 0 & -\sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ \sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Ux rotates x through angle θ in the $(1,3)$-coordinate plane.

(2) Householder transformations

Consider <u>nonzero</u> $w \in \mathbb{C}^n$

Form $\boxed{U_w = I - t w w^*, \quad t = \frac{2}{w^* w}}$

Then:

$$U_w^* U_w = (I - t w w^*)^* (I - t w w^*)$$

$$= I - t w w^* - t w w^* + t^2 w w^* w w^*$$

$$= I - 2 t w w^* + \frac{2}{w^* w} \frac{2}{\cancel{w^* w}} w \cancel{(w^* w)} w^*$$

$$= I - 4 \frac{w w^*}{w^* w} + \frac{4}{w^* w} w w^* = I$$

so indeed U_w is unitary.

<u>Facts</u>

- U_w is hermitian $(U_w^* = U_w)$
- If $x \perp w$ $(x^* w = 0)$, then

$$U_w x = (I - t w w^*) x = x - 0 = x$$

i.e. U_w is <u>the identity</u> on $\{w\}^\perp$

- $U_w w = w - t w w^* w = w - 2w = -w$

i.e. U_w is a reflection of span$\{w\}$.

Notation $\{\ \}^{\perp}$: orthogonal complement

$$\{w\}^{\perp} = \{x \in \mathbb{C}^n : x \perp w\}$$
$$= \{x \in \mathbb{C}^n : x^* w = 0\}.$$

<u>Exercise</u> Prove that $\{w\}^{\perp}$ is a subspace.

- $\{0\}^{\perp} = \mathbb{C}^n$
- For any subspace $W \subseteq \mathbb{C}^n$,

$$W^{\perp} = \{x \in \mathbb{C}^n : x^* w = 0, \forall w \in W\}$$

orthogonal complement of a vector space

<u>Theorem</u> (Orthogonal Decomposition Theorem)

Given any $x \in \mathbb{C}^n$ and a subspace $W \subseteq \mathbb{C}^n$, we can uniquely write

$$x = y + z$$

where $y \in W$, $z \in W^{\perp}$.

<u>Example</u>

$$w \in \mathbb{C}^n, \quad w \neq 0$$

$$W = \text{span}\{w\}$$

Orthogonal Decomp. Theorem

$\forall x \in \mathbb{C}^n$, write

$$x = y + z, \quad y = \alpha w \in W, \alpha \in \mathbb{C}$$
$$z \perp w \quad (z \in W^{\perp})$$

Then

$$U_w x = W_w y + V_w z$$
$$= -y + z = z - y.$$

Predicable action on every vector.

§ 2.3 Schur's Unitary Triangularization Theorem (Schur's Theorem)

Most fundamental fact!

MacTutor
↓
check bio of Schur

<u>Theorem</u> (Schur)

Given any $A \in M_n$ with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (in any order), there exist unitary $U \in M_n$ such that

$$U^* A U = T = [t_{ij}]$$

is upper triangular (i.e. $t_{ij} = 0$ if $i > j$)

with $t_{ii} = \lambda_i$, $i = 1, 2, \ldots, n$

(If $A \in M_n(\mathbb{R})$, $\sigma(A) \subseteq \mathbb{R}$, U can be taken to be "real orthogonal": $U^* = U^T = U^{-1}$)

Observation: $U^* A U$ is a similarity!

T being triangular, we have

$$t_{ii} \in \sigma(T)$$

$$\sigma(T) = \sigma(A) = \{t_{11}, t_{22}, \ldots, t_{nn}\}$$
$$= \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$$